

QuickCell-12 Parameter Guide

2-Input & 3-Input Gate Reference

command-line parameter reference for filter calibration

This document lists references for all the tunable parameters in the QuickCell-12 prototype. This includes both the 2- input gate design ($d = 3$ canvas SiDBs, $\mu_- = -0.32$ eV) and 3 input gate designs ($d = 4$ canvas SiDBs, $\mu_- = -0.31$ eV) and each tunable parameter corresponds to (Pruning filter, increasing/decreasing effect, and useful range)

A number of tunable parameters in the prototype represent physical margins, while the rest of them refer to robustness parameters.

There are several calibrated pruning margins that help filters to work either in the permissive physical feasibility mode or in the restrictive robustness-oriented mode. smaller or larger value does not necessarily imply anything better. Instead, such parameters define the tradeoff between the candidate set size and pruning aggressiveness. Permissive modes should be applied for 2-input gates due to their small canvas and geometrical constraints, but 3-input gates support more aggressive filter settings

Gate Overview

2-Input Gates (15 gates)

Category	Gate Names	Inputs	Outputs	d	μ_- [eV]	Script
2-in, 1-out	AND, NAND, OR, NOR, XOR, XNOR, LT, GT, LE, GE	2	1	3	-0.32	run_gate_report_2input.py
2-in, 2-out	DOUBLE WIRE, CX, HA	2	2	3	-0.32	run_gate_report_2input.py
1-in, 1-out	WIRE, INV	1	1	3	-0.32	run_gate_report_2input.py

3-Input Gates (10 gates)

Category	Gate Names	Inputs	Outputs	d	μ_- [eV]	Script
3-in, 1-out	AND3, XOR-AND, OR-AND, ONEHOT, MAJ, GAMBLE, DOT, ITE, AND-XOR, XOR3	3	1	4	-0.31	run_gate_report.py

Key Differences Between 2-Input and 3-Input Setups

Parameter	2-Input Gates	3-Input Gates
μ_- (mu minus)	-0.32 eV	-0.31 eV
Canvas SiDBs (d)	3	4
Input patterns	4 (2^2) or 2 (1^1)	8 (2^3)
Skeleton (1-out)	17 SiDBs	25 SiDBs
Canvas size (1-out)	99 positions	143 positions
$ L_d $ (1-out)	$C(99,3) = 156,849$	$C(143,4) = 17,058,156$
Total gates	15	10
Report script	run_gate_report_2input.py	run_gate_report.py
Export script	export_sqad_validation_2input.py	export_sqad_validation.py
Results folder	results_2input/	results/

Example Commands — 3-Input Gates

Balanced Setting (recommended for paper)

```
python run_gate_report.py --gate AND3 --samples 100000 --workers 4 --chunksize 2000 --strict-profile balanced
--io-margin 0.0001 --connectivity-radius 9.0 --d-min 3.1 --positive-charge-margin 0.02 --charge-min-fraction
0.25 --charge-max-fraction 1.0 --charge-count-conflict-radius 4.3 --input-disturbance 0.001 --pressure-margin
0.001 --energy-margin 0.01 --fll-mode strict-hop --fll-min-support-states 2
```

Run ALL 3-input gates

```
python run_gate_report.py --gate ALL --samples 100000 --workers 4 --chunksize 2000 --strict-profile balanced --io-margin 0.0001
```

Export SiQAD files — 3-input

```
python export_siqa_validation.py --gate ALL --samples 100000 --workers 4 --chunksize 2000 --strict-profile balanced --io-margin 0.0001 --max-candidates 50
```

Example Commands — 2-Input Gates

Balanced Setting (easy gates)

```
python run_gate_report_2input.py --gate AND --samples 200000 --workers 4 --strict-profile safe --io-margin 0.02
```

Run ALL 2-input gates (maximum success)

```
python run_gate_report_2input.py --gate ALL --samples 1000000 --workers 4 --chunksize 5000 --canvas-d 3 --seed 12345 --sim-time 0.01 --strict-profile safe --io-margin 0.05 --pressure-margin 0.0 --d-min 1.0 --positive-charge-margin 0.0 --charge-min-fraction 0.0 --charge-max-fraction 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --energy-margin 1.0 --f11-mode counted --f11-min-support-states 1 --selection-target 20 --selection-min-margin 0.0
```

Difficult gates (XOR, XNOR, CX, HA)

```
python run_gate_report_2input.py --gate XOR --samples 1000000 --workers 4 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

Export SiQAD files — ALL 2-input gates

```
python export_siqa_validation_2input.py --gate ALL --samples 1000000 --workers 4 --chunksize 5000 --max-candidates 20 --canvas-d 3 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --energy-margin 1.0 --f11-mode counted --selection-target 20 --selection-min-margin 0.0
```

Export individual 2-input gates

AND / NAND / OR / NOR / LE / GE / WIRE / INV (easy)

```
python export_siqa_validation_2input.py --gate AND --samples 200000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.02
```

XOR (hardest 2-input gate)

```
python export_siqa_validation_2input.py --gate XOR --samples 1000000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

XNOR

```
python export_siqa_validation_2input.py --gate XNOR --samples 1000000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

LT ($A < B$)

```
python export_siqa_validation_2input.py --gate LT --samples 500000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

GT ($A > B$)

```
python export_siqa_validation_2input.py --gate GT --samples 500000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

DOUBLE_WIRE (2-in, 2-out)

```
python export_siqa_validation_2input.py --gate DOUBLE_WIRE --samples 500000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

CX — Crossover (2-in, 2-out)

```
python export_siqa_validation_2input.py --gate CX --samples 1000000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

HA — Half Adder (2-in, 2-out)

```
python export_siqa_validation_2input.py --gate HA --samples 1000000 --workers 4 --max-candidates 20 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --f11-mode counted
```

Analyze SiQAD results

```
# 3-input gates
python analyze_siqa_results.py --root validation_exports --all
# 2-input gates
python analyze_siqa_results.py --root validation_exports_2input --all
```

Complete Parameter Summary Table

this table lists every tunable parameters (the filter it controls, meaning, effect of smaller/larger values, recommended ranges for both 2-input and 3-input gate setups)

Parameter	Filter	Meaning	Smaller Value	Larger Value	Range (3-in)	Range (2-in)	Current (3-in)	Current (2-in)
--d-min	F1	Minimum distance between canvas SiDBs	Weaker F1	Stricter F1	3.0 – 4.5	1.0 – 4.5	3.1	1.0
--positive-charge-margin	F4	Safety margin for positive-charge activation	Weaker F4	Stricter F4	0.005 – 0.05	0.0 – 0.05	0.02	0.0
--charge-min-fraction	F5	Minimum required negative-charge fraction	Weaker F5	Stricter F5	0.20 – 0.28	0.0 – 0.25	0.25	0.0
--charge-max-fraction	F5	Maximum allowed negative-charge fraction	Stricter if reduced	Weaker if increased	0.8 – 1.0	0.8 – 1.0	1.0	1.0
--charge-count-conflict-radius	F5	Conflict radius for simultaneous negative canvas occupation	Weaker F5	Stricter F5	4.0 – 6.5	None – 6.5	4.3	None
--input-disturbance	F6	Maximum canvas-induced disturbance on input pins	Stricter F6	Weaker F6	0.001 – 0.01	0.3 – 1.0	0.001	1.0
--connectivity-radius	F7	Electrostatic connectivity radius for path check	Stricter F7	Weaker F7	8.5 – 9.8	15.0 – 25.0	9.0	25.0
--pressure-margin	F9	Required safety margin for output BDL pressure	Weaker F9	Stricter F9	0.0001 – 0.002	0.0	0.001	0.0
--energy-margin	F10	Energy margin for relaxed energetic pruning	Stricter F10	Weaker F10	0.005 – 0.05	0.5 – 1.0	0.01	1.0
--f11-mode	F11	Physical infeasibility execution mode	N/A	N/A	counted / strict-pop / strict-hop	counted / strict-pop / strict-hop	strict-hop	counted
--f11-min-support-states	F11	Min metastable supporting states per input	Weaker F11	Stricter F11	1 – 3	1	2	1
--io-margin	F12	Energy tolerance for I/O instability pruning	Stricter F12	Weaker F12	0.00001 – 0.002	0.02 – 0.05	0.0001	0.05
--strict-profile	Multiple	Preset profile: safe/balanced/strict	N/A	N/A	Safe, balanced, strict	safe, balanced, strict	balanced	safe
--samples	N/A	Number of random layout samples to evaluate	Faster but may miss candidates	Slower but more thorough	100,000 – 500,000	200,000 – 1,000,000	100,000	1,000,000
--canvas-d	N/A	Number of canvas SiDBs per layout	N/A	N/A	4 (fixed)	3 (fixed)	4	3
--workers	N/A	Number of parallel worker processes	N/A	N/A	1 – 8	1 – 8	4	4
--chunksize	N/A	Layouts per work chunk	N/A	N/A	1000 – 5000	1000 – 5000	2000	5000
--seed	N/A	Random seed for reproducibility	N/A	N/A	Any integer	Any integer	12345	12345
--sim-time	N/A	Assumed per-layout simulation time for speedup calc	N/A	N/A	0.01	0.01	0.01	0.01
--selection-target	Post-F12	Top-K energy-margin ranking after F12	N/A	N/A	0 (disabled) or 20–50	0 or 20–50	0	20
--selection-min-margin	Post-F12	Minimum energy margin for exported candidates	N/A	N/A	None or 0.001	None or 0.0	None	0.0

Filter-by-Filter Parameter Explanation

F1 — Minimum Distance Pruning

--d-min :

F1 remove a layout , if at least two selected canvas SiDBs are closer than $d_{\min}$. in the rent paper like canvas, the grid spacing is approximately 3.0 nm. For 3-input gates, using --d-min 3.1 introduces a design rule spacing threshold. For 2-input gates with difficult truth tables, using --d-min 1.0 allows more layouts to survive and increases the chance of finding valid candidates

Value	Expected Effect	Recommended For
1.0	Very permissive , almost no pruning	2-input difficult gates (XOR, CX, HA)
3.1	Removes immediate neighboring placements	3-input gates (standard)
4.3	Also removes diagonal neighbors ($\sqrt{3^2+3^2} \approx 4.24$)	3-input aggressive
> 5.0	Very strict , may remove most layouts	Rarely useful

F4 — Positive Charge Pruning

--positive-charge-margin :

F4 removes layouts that can enter a positive charge regime , A margin of 0.0 gives the original behavior and for 2-input gates that struggle to find candidates, keep this at 0.0

Value	Expected Effect
0.0	Original F4 behavior (recommended for 2-input)
0.005	Mild additional pruning
0.01 – 0.02	Moderate pruning (recommended for 3-input balanced)
0.05	Strong pruning; possible over-pruning

F5 — Charge Count Bound Pruning

F5 uses charge capacity constraints. Three sub parameters control it:

--charge-min-fraction: minimum required negative charge fraction

--charge-max-fraction: maximum allowed negative charge fraction

--charge-count-conflict-radius: conflict distance for simultaneous negative occupation

Parameter	3-Input Range	2-Input Range
--charge-min-fraction	0.20 – 0.28	0.0 (permissive)
--charge-max-fraction	0.8 – 1.0	1.0
--charge-count-conflict-radius	4.0 – 6.5	None (disabled)

F6 — Input Pin Disturbance Pruning

--input-disturbance :

F6 protects input BDL pins from excessive electrostatic back action , for 2-input difficult gates, a very large value (e.g 1.0) effectively disables this filter to preserve candidates

Value	Expected Effect
None	Uses default threshold; may prune little
1.0	Effectively disables F6 (recommended for 2-input difficult)
0.01	Mild pruning
0.003	Moderate pruning
0.001	Very strict (recommended for 3-input balanced)

F7 — Electrostatic Path Connectivity Pruning

--connectivity-radius :

F7 constructs a connectivity graph, the layout is rejected if input/output reachability is not satisfied within the radius. for 2-input gates, a large radius (25.0) effectively loosens this filter

Value	Context	Expected Effect
5.0	Any	It is too strict , removes almost everything
8.5 – 9.0	3-input	Useful pruning region
9.5 – 10.0	3-input	Milder pruning
15.0 – 25.0	2-input	Permissive, preserves candidates
25.0	2-input difficult	Effectively minimal pruning

F8 — Output Potential Bound Pruning

F8 checks whether the output BDL sites can sustain the required charge states given the potential interval from skeleton + canvas , this filter has no explicit tunable margin parameter , it uses the physical charge stability windows from μ_- and μ_+

F9 — Output Pressure Pruning

--pressure-margin :

F9 checks whether the output BDL pair can be biased in the required direction, for 2-input gates, keeping this at 0.0 is recommended

Value	Expected Effect
0.0	Basic sign check only (recommended for 2-input)
0.0005	Moderate pruning
0.001	Stronger pruning (recommended for 3-input balanced)
0.002	Very strict , may remove many candidates

F10 — Relaxed Energy Pruning

--energy-margin :

F10 compares relaxed minimum energies, If an incorrect assignment is lower by more than energy margin, the layout will be removed

Smaller = stricter, Larger = weaker

For 2 input difficult gates, a large value (1.0) effectively disables F10

Value	Expected Effect
1.0	It is very weak, almost no pruning (2-input difficult)
0.05	Weak(F10)
0.01	Balanced (3-input recommended)
0.005	Strong pruning
0.001	Very strict

F11 — Physical Infeasibility Pruning

--f11-mode and --f11-min-support-states

F11 checked whether the correct I/O can be physically supported by at least one metastable canvas charge configuration

--f11-mode	Meaning	Recommended For
counted	F11 is not executed , counted as passed	2 input gates (maximize candidates)
strict-pop	Population stability is checked	3-input conservative
strict-hop	Population + single-electron hop stability	3 input balanced/aggressive

--f11-min-support-states	Expected Effect	Recommended
1	Standard feasibility	2-input gates, 3-input conservative
2	More robust	3 input balanced
3	Stronger robustness	3 input aggressive
> 4	May be too strict for d = 3 or d = 4	Rarely useful

F12 — I/O Signal Instability Pruning

--io-margin

F12 is the final and most critical filter. this checks whether an incorrect output assignment has lower energy than the correct one
smaller = stricter, larger = weaker.

this is most important parameter for getting candidates, If no candidates are found, increase this value first

Value	Expected Effect	Recommended For
0.05	Very permissive	2 input difficult gates (XOR, CX, HA)
0.02	Permissive	2 input easy gates
0.001	Moderate	3-input conservative
0.0001	Strict	3 input balanced
0.00001	Very strict	3 input aggressive

Recommended Parameter Sets

3-Input Gates

Conservative / Candidate-Preserving (3-input)

```
--strict-profile balanced --d-min 3.1 --positive-charge-margin 0.01 --charge-min-fraction 0.20 --charge-max-fraction 1.0 --charge-count-conflict-radius 4.3 --input-disturbance 0.003 --connectivity-radius 9.5 --pressure-margin 0.0005 --energy-margin 0.02 --f11-mode strict-pop --f11-min-support-states 1 --io-margin 0.0005
```

Balanced (3-input) — Recommended for paper

```
--strict-profile balanced --d-min 3.1 --positive-charge-margin 0.02 --charge-min-fraction 0.25 --charge-max-fraction 1.0 --charge-count-conflict-radius 4.3 --input-disturbance 0.001 --connectivity-radius 9.0 --pressure-margin 0.001 --energy-margin 0.01 --f11-mode strict-hop --f11-min-support-states 2 --io-margin 0.0001
```

Aggressive Pruning (3-input)

```
--strict-profile balanced --d-min 4.3 --positive-charge-margin 0.03 --charge-min-fraction 0.27 --charge-max-fraction 1.0 --charge-count-conflict-radius 6.1 --input-disturbance 0.0008 --connectivity-radius 8.7 --pressure-margin 0.002 --energy-margin 0.005 --f11-mode strict-hop --f11-min-support-states 3 --io-margin 0.00005
```

2-Input Gates

Easy gates (AND, OR, NAND, NOR, LE, GE, WIRE, INV)

```
--strict-profile safe --samples 200000 --io-margin 0.02 --d-min 1.0 --positive-charge-margin 0.0 --charge-min-fraction 0.0 --charge-max-fraction 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --pressure-margin 0.0 --energy-margin 1.0 --f11-mode counted --f11-min-support-states 1
```

Difficult gates (XOR, XNOR, LT, GT, DOUBLE_WIRE, CX, HA)

```
--strict-profile safe --samples 1000000 --io-margin 0.05 --d-min 1.0 --positive-charge-margin 0.0 --charge-min-fraction 0.0 --charge-max-fraction 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --pressure-margin 0.0 --energy-margin 1.0 --f11-mode counted --f11-min-support-states 1
```

Maximum success (ALL 2-input gates)

```
--strict-profile safe --samples 1000000 --chunksize 5000 --canvas-d 3 --seed 12345 --io-margin 0.05 --pressure-margin 0.0 --d-min 1.0 --positive-charge-margin 0.0 --charge-min-fraction 0.0 --charge-max-fraction 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --energy-margin 1.0 --f11-mode counted --f11-min-support-states 1 --selection-target 20 --selection-min-margin 0.0
```

Troubleshooting: No Candidates Found

If a gate produce zero candidates, follow these steps in order:

Step	Action	Why
1	Increase (--io-margin to 0.05)	F12 is the strictest filter
2	Change (--strict-profile to safe)	Relaxes wire radius and d_min
3	Increase (--samples to 1000000)	More layouts explored
4	Set (--f11-mode counted)	Disables strict F11
5	Set (--input-disturbance 1.0)	Effectively disables F6
6	Set (--connectivity-radius 25.0)	Effectively disables F7
7	Set (--energy-margin 1.0)	Effectively disables F10
8	Set (--d-min 1.0)	Effectively disables F1
9	Set (--positive-charge-margin 0.0)	Original F4 behavior

If no candidates are found, the first parameter to relax is always --io-margin, followed by --strict-profile safe, then --samples increase. The remaining filter parameters should only be relaxed if the first three steps are insufficient

Gate Difficulty Classification

2-Input Gates

Difficulty	Gates	Recommended --samples	Recommended --io-margin
Easy	AND, NAND, OR, NOR, LE, GE, WIRE, INV	200,000	0.02
Medium	LT, GT, DOUBLE WIRE	500,000	0.05
Hard	XOR, XNOR, CX, HA	1,000,000	0.05

3-Input Gates

Difficulty	Gates	Recommended --samples	Recommended --io-margin
Easy	AND3, OR-AND, MAJ	100,000	0.0001
Medium	XOR-AND, ONEHOT, GAMBLE, ITE	100,000	0.0001
Hard	DOT, AND-XOR, XOR3	200,000	0.0005

Filter Type Classification

Filter	Type	Tunable Parameter
F1	Design-rule / geometric pruning	--d-min
F2	Symmetry-based canonical pruning	(automatic)
F3	Wire interference geometric pruning	--wire-radius (via profile)
F4	Physical positive-charge guard	--positive-charge-margin
F5	Heuristic charge-capacity pruning	--charge-min/max-fraction, --charge-count-conflict-radius
F6	Heuristic input-integrity pruning	--input-disturbance
F7	Heuristic electrostatic reachability	--connectivity-radius
F8	Analytical output-potential feasibility	(no explicit margin)
F9	Analytical output-pressure feasibility	--pressure-margin
F10	Relaxed energetic pre-pruning	--energy-margin
F11	Enumerative physical feasibility	--f11-mode, --f11-min-support-states
F12	Enumerative I/O energetic stability	--io-margin

Complete Command Reference

Full 3-input report command

```
python run_gate_report.py --gate ALL --samples 100000 --workers 4 --chunksize 2000 --seed 12345 --sim-time 0.01 --strict-profile balanced --io-margin 0.0001 --pressure-margin 0.001 --connectivity-radius 9.0 --d-min 3.1 --positive-charge-margin 0.02 --charge-min-fraction 0.25 --charge-max-fraction 1.0 --charge-count-conflict-radius 4.3 --input-disturbance 0.001 --energy-margin 0.01 --f11-mode strict-hop --f11-min-support-states 2 --selection-target 0
```

Full 2-input report command

```
python run_gate_report_2input.py --gate ALL --samples 1000000 --workers 4 --chunksize 5000 --canvas-d 3 --seed 12345 --sim-time 0.01 --strict-profile safe --io-margin 0.05 --pressure-margin 0.0 --connectivity-radius 25.0 --d-min 1.0 --positive-charge-margin 0.0 --charge-min-fraction 0.0 --charge-max-fraction 1.0 --input-disturbance 1.0 --energy-margin 1.0 --fll-mode counted --fll-min-support-states 1 --selection-target 20 --selection-min-margin 0.0
```

Full 3-input SiQAD export command

```
python export_siqad_validation.py --gate ALL --samples 100000 --workers 4 --chunksize 2000 --max-candidates 50 --strict-profile balanced --io-margin 0.0001
```

Full 2-input SiQAD export command

```
python export_siqad_validation_2input.py --gate ALL --samples 1000000 --workers 4 --chunksize 5000 --max-candidates 20 --canvas-d 3 --strict-profile safe --io-margin 0.05 --d-min 1.0 --input-disturbance 1.0 --connectivity-radius 25.0 --energy-margin 1.0 --fll-mode counted --selection-target 20 --selection-min-margin 0.0
```

Truth Tables Reference

2-Input, 1-Output Gates

Gate	Hex	00	01	10	11
AND	0x8	0	0	0	1
NAND	0x7	1	1	1	0
OR	0xE	0	1	1	1
NOR	0x1	1	0	0	0
XOR	0x6	0	1	1	0
XNOR	0x9	1	0	0	1
LT	0x2	0	1	0	0
GT	0x4	0	0	1	0
LE	0xB	1	1	0	1
GE	0xD	1	0	1	1

2-Input, 2-Output Gates

Gate	Hex (O ₁ ,O ₂)	00→	01→	10→	11→
DOUBLE WIRE	(0xC, 0xA)	0,0	0,1	1,0	1,1
CX	(0xA, 0xC)	0,0	1,0	0,1	1,1
HA	(0x6, 0x8)	0,0	1,0	1,0	0,1

1-Input, 1-Output Gates

Gate	Hex	0	1
WIRE	0x2	0	1
INV	0x1	1	0

3-Input, 1-Output Gates

Gate	Hex	000	001	010	011	100	101	110	111
AND3	0x80	0	0	0	0	0	0	0	1
XOR-AND	0x28	0	0	0	1	0	1	0	0
OR-AND	0xA8	0	0	0	1	0	1	0	1
ONEHOT	0x16	0	1	1	0	1	0	0	0
MAJ	0xE8	0	0	0	1	0	1	1	1
GAMBLE	0x81	1	0	0	0	0	0	0	1
DOT	0x52	0	1	0	0	1	0	1	0
ITE	0xD8	0	0	0	1	1	1	0	1
AND-XOR	0x6A	0	1	0	1	0	1	1	0
XOR3	0x96	0	1	1	0	1	0	0	1